

IBM BOB 2.0 HACKATHON

Legacy Bridge

IBM Bob modernizes legacy Java.

Bob's own tests decide what ships.

THE PROBLEM

**Nobody knows
everything old
code does.**

AI refactors can change behaviour
while making code look cleaner.
Nobody can review every edge case
of a 300-line diff.

```
* Copyright 2001-2004 The Apache Software  
Foundation.
```

```
...
```

```
public class Base64 implements BinaryEncoder,  
BinaryDecoder {
```

```
    static final int CHUNK_SIZE = 76;
```

```
    static final byte[] CHUNK_SEPARATOR =  
"\r\n".getBytes();
```

```
...
```

```
    public static byte[] decodeBase64(byte[]  
base64Data) {
```

```
        // RFC 2045 requires that we discard ALL  
non-Base64 characters
```

```
        base64Data = discardNonBase64(base64Data);
```

Real code: Apache Commons Codec 1.3 · Base64 · 2005

HOW IT WORKS

legacy class → Bob writes characterization tests → must pass on the old code

|



mutation check → trap slipped through? Bob hardens its tests

|



Bob modernizes from a ticket → re-run Bob's tests

└─ all pass: VERIFIED, applied

└─ any fail: BLOCKED, back to Bob to repair (same gate)

RESULTS · REAL BOB RUNS

Code	Ticket	Bob's tests	Outcome
Base64 Apache Commons Codec, 2005	Reimplement on java.util.Base64	58 / 60 · 5/5 traps	BLOCKED 2 of 2 runs even a bad Bob repair was refused
Metaphone Apache Commons Codec, 2005	Refactor the 170-line switch	117 · 4/5 → 5/5 traps	VERIFIED 0 of 136,800 outputs changed
Invoice teaching example	Simplify the branching logic	61 · 4/4 traps	BLOCKED → repaired \$25.00 would have become \$23.75

19 real Bob calls · 2.54 Bob coins · all recorded in bob_sessions/

WHY THIS IS IBM BOB AT WORK

Bob does every reasoning step.

- 1 · Writes the characterization tests (used its own subagents on real code)
- 2 · Fixes its own wrong expectations
- 3 · Hardens its suite when a mutant survives
- 4 · Modernizes from a real ticket
- 5 · Repairs blocked changes, which must pass the same gate

The harness only builds, runs tests, applies mutants and decides VERIFIED or BLOCKED. It never edits code.

WHAT'S NEXT

Every AI change should come with proof.

- Run the gate inside Bob as a skill, on any repo
- More languages: COBOL and RPG on IBM i, where legacy risk is highest
- CI integration: no AI modernization merges without its characterization proof
- Differential checks for every target, not just Metaphone